

Book My Show

Brief

Bookmyshow is a ticketing platform where you can book tickets for a movie show. The image attached represents that for a given theatre we can see the next 7 dates. As one chooses the date, we get a list of all shows running in that theatre along with the show timings.

P1 - As part of this assignment, we need to list down all the entities, their attributes and the table structures for the scenario mentioned in the previous slide. You also need to write the SQL queries required to create these tables along with a few sample entries. Ensure the tables follow 1NF, 2NF, 3NF and BCNF rules.

P2 - Write a query to list down all the shows on a given date at a given theatre along with their respective show timings.



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Entities

1. City
 - a. Entity Type: City
 - b. Entity Set: cities
 - c. Attributes:
 - i. id (Primary Key)
 - ii. name
 - iii. state
 - iv. country
2. Theater Brand
 - a. Entity Type: TheaterBrand
 - b. Entity Set: theater_brands
 - c. Attributes:
 - i. id (Primary Key)
 - ii. name
3. Theater Branch
 - a. Entity Type: Theater
 - b. Entity Set: theaters
 - c. Attributes:
 - i. id (Primary Key)
 - ii. brand_id (Foreign Key)
 - iii. city_id (Foreign Key)
 - iv. branch_name
 - v. address
 - vi. latitude
 - vii. longitude
4. Screen
 - a. Entity Type: Screen
 - b. Entity Set: screens
 - c. Attributes:
 - i. id (Primary Key)
 - ii. theater_id (Foreign Key)
 - iii. screen_number
 - iv. capacity
5. Movie
 - a. Entity Type: Movie
 - b. Entity Set: movies
 - c. Attributes:
 - i. id (Primary Key)
 - ii. title
 - iii. duration_min

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6. Language
 - a. Entity Type: Language
 - b. Entity Set: languages
 - c. Attributes:
 - i. id (Primary Key)
 - ii. name
7. Format(2D / IMAX etc.)
 - a. Entity Type: Format
 - b. Entity Set: formats
 - c. Attributes:
 - i. id (Primary Key)
 - ii. name
8. Audio Format
 - a. Entity Type: AudioFormat
 - b. Entity Set: audio_formats
 - c. Attributes:
 - i. id (Primary Key)
 - ii. name
9. Show
 - a. Entity Type: Show
 - b. Entity Set: shows
 - c. Attributes:
 - i. id (Primary Key)
 - ii. screen_id (Foreign Key)
 - iii. movie_id (Foreign Key)
 - iv. language_id (Foreign Key)
 - v. format_id (Foreign Key)
 - vi. audio_format_id (Foreign Key)
 - vii. show_time

Queries

-- 1. City Table (One City -> Many Theaters)

```
CREATE TABLE cities (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100) NOT NULL,  
  state VARCHAR(100),  
  country VARCHAR(100)  
);
```

-- 2. Theater Brands Table

```
CREATE TABLE theater_brands (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(255) NOT NULL  
);
```

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-- 3. Theaters Table

```
CREATE TABLE theaters (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  brand_id INT NOT NULL,  
  city_id INT NOT NULL,  
  branch_name VARCHAR(255), -- e.g. Nexus Mall  
  address TEXT,  
  latitude DECIMAL(10,8),  
  longitude DECIMAL(11,8),  
  
  FOREIGN KEY (brand_id) REFERENCES theater_brands(id),  
  FOREIGN KEY (city_id) REFERENCES cities(id)  
);
```

-- 4. Audio Formats Table

```
CREATE TABLE audio_formats (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100) NOT NULL  
);
```

-- 5. Screens Table (One Theater -> Many Screens)

```
CREATE TABLE screens (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  theater_id INT NOT NULL,  
  screen_number INT NOT NULL,  
  capacity INT,  
  FOREIGN KEY (theater_id) REFERENCES theaters(id) ON DELETE CASCADE  
) ENGINE=InnoDB;
```

-- 6. Movies Table

```
CREATE TABLE movies (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  title VARCHAR(255) NOT NULL,  
  duration_min INT  
) ENGINE=InnoDB;
```

-- 7. Language Table

```
CREATE TABLE languages (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100) NOT NULL  
);
```

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-- 8. formats Table

```
CREATE TABLE formats (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(50) NOT NULL  
);
```

-- 9. Showtimes Table (The Junction)

-- Connects a movie to a screen at a specific date/time

```
CREATE TABLE shows (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  screen_id INT NOT NULL,  
  movie_id INT NOT NULL,  
  language_id INT NOT NULL,  
  format_id INT NOT NULL,  
  audio_format_id INT NOT NULL,  
  show_time DATETIME NOT NULL,  
  FOREIGN KEY (screen_id) REFERENCES screens(id) ON DELETE CASCADE,  
  FOREIGN KEY (movie_id) REFERENCES movies(id) ON DELETE CASCADE,  
  FOREIGN KEY (language_id) REFERENCES languages(id) ON DELETE CASCADE,  
  FOREIGN KEY (format_id) REFERENCES formats(id) ON DELETE CASCADE,  
  FOREIGN KEY (audio_format_id) REFERENCES audio_formats(id) ON DELETE CASCADE,  
  
  -- Constraint: A screen cannot show two movies at the exact same time  
  UNIQUE KEY unique_screening (screen_id, show_time)  
) ENGINE=InnoDB;
```

-- Inserting dummy Data

```
INSERT INTO cities (name, state, country) VALUES  
( 'Bangalore', 'Karnataka', 'India'),  
( 'Mumbai', 'Maharashtra', 'India'),  
( 'Hyderabad', 'Telangana', 'India');
```

```
INSERT INTO theater_brands (name) VALUES  
( 'PVR Cinemas'),  
( 'INOX'),  
( 'Cinepolis');
```

```
INSERT INTO theaters  
(brand_id, city_id, branch_name, address, latitude, longitude)  
VALUES  
-- Bangalore  
(1, 1, 'Nexus Mall', 'Koramangala, Bangalore', 12.9352, 77.6245),
```

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(2, 1, 'Forum Mall', 'Whitefield, Bangalore', 12.9698, 77.7500),

-- Mumbai

(1, 2, 'Phoenix Mall', 'Lower Parel, Mumbai', 19.0023, 72.8258),

-- Hyderabad

(3, 3, 'Inorbit Mall', 'Hitech City, Hyderabad', 17.4340, 78.3866);

INSERT INTO audio_formats (name) VALUES

('Dolby Atmos'),

('Dolby 7.1'),

('DTS'),

('Auro 3D'),

('Stereo');

INSERT INTO screens (theater_id, screen_number, capacity) VALUES

-- Nexus Mall (Bangalore)

(1, 1, 200),

(1, 2, 150),

-- Forum Mall (Bangalore)

(2, 1, 180),

-- Phoenix Mall (Mumbai)

(3, 1, 220),

-- Inorbit Mall (Hyderabad)

(4, 1, 190);

INSERT INTO movies (title, duration_min) VALUES

('Avengers: Final War', 160),

('Dune: Part Three', 155),

('KGF Chapter 3', 170),

('Salaar 2', 165);

INSERT INTO languages (name) VALUES

('English'),

('Hindi'),

('Kannada'),

('Telugu'),

('Tamil');

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```
INSERT INTO formats (name) VALUES
```

```
('2D'),  
('3D'),  
('IMAX'),  
('4DX');
```

```
INSERT INTO shows (screen_id, movie_id, language_id, format_id, show_time,  
audio_format_id) VALUES
```

```
-- Bangalore - Nexus Mall Screen 1  
(1, 1, 1, 2, '2026-02-22 10:00:00', 1), -- Avengers English 3D  
(1, 2, 1, 3, '2026-02-22 14:00:00', 3), -- Dune English IMAX  
(1, 3, 3, 1, '2026-02-22 18:00:00', 2), -- KGF Kannada 2D
```

```
-- Bangalore - Nexus Mall Screen 2  
(2, 1, 2, 1, '2026-02-22 11:00:00', 1), -- Avengers Hindi 2D  
(2, 4, 4, 1, '2026-02-22 15:00:00', 4), -- Salaar Telugu 2D
```

```
-- Bangalore - Forum Mall  
(3, 3, 4, 1, '2026-02-22 12:00:00', 5), -- KGF Telugu 2D
```

```
-- Mumbai - Phoenix Mall  
(4, 1, 1, 2, '2026-02-22 10:30:00', 2), -- Avengers English 3D  
(4, 2, 2, 2, '2026-02-22 17:00:00', 5), -- Dune Hindi 3D
```

```
-- Hyderabad - Inorbit Mall  
(5, 4, 4, 1, '2026-02-22 13:00:00', 3), -- Salaar Telugu 2D  
(5, 3, 3, 1, '2026-02-22 19:00:00', 4); -- KGF Kannada 2D
```

```
-- Querying the different Tables
```

```
SELECT * FROM cities;  
SELECT * FROM theater_brands;  
SELECT * FROM theaters;  
SELECT * FROM audio_formats;  
SELECT * FROM screens;  
SELECT * FROM movies;  
SELECT * FROM languages;  
SELECT * FROM formats;  
SELECT * FROM shows;
```

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```
SELECT
  tb.name AS brand,
  t.branch_name,
  m.title,
  s.show_time,
  l.name AS language,
  f.name AS videoFormat,
  af.name AS audioFormat,
  c.name AS city
FROM shows s
JOIN screens sc ON s.screen_id = sc.id
JOIN theaters t ON sc.theater_id = t.id
JOIN theater_brands tb ON t.brand_id = tb.id
JOIN cities c ON t.city_id = c.id
JOIN movies m ON s.movie_id = m.id
JOIN languages l ON s.language_id = l.id
JOIN formats f ON s.format_id = f.id
JOIN audio_formats af ON s.audio_format_id = af.id
WHERE c.name = 'Bangalore' AND tb.name = 'PVR Cinemas'
AND s.show_time >= '2026-02-22 00:00:00'
AND s.show_time < '2026-02-23 00:00:00'
ORDER BY s.show_time;
```

Output:

-- 1. City Table

id	name	state	country
1	Bangalore	Karnataka	India
2	Mumbai	Maharashtra	India
3	Hyderabad	Telangana	India

-- 2. Brand Table

id	name
1	PVR Cinemas
2	INOX
3	Cinepolis

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-- 3. Theater Table

id	brand_id	city_id	branch_name	address	latitude	longitude
1	1	1	Nexus Mall	Koramangala, Bangalore	12.93520000	77.62450000
2	2	1	Forum Mall	Whitefield, Bangalore	12.96980000	77.75000000
3	1	2	Phoenix Mall	Lower Parel, Mumbai	19.00230000	72.82580000
4	3	3	Inorbit Mall	Hitech City, Hyderabad	17.43400000	78.38660000

-- 4. Audio Formats Table

id	name
1	Dolby Atmos
2	Dolby 7.1
3	DTS
4	Auro 3D
5	Stereo

-- 5. Screens Table

id	theater_id	screen_number	capacity
1	1	1	200
2	1	2	150
3	2	1	180
4	3	1	220
5	4	1	190

-- 6. Movies Table

id	title	duration_min
1	Avengers: Final War	160
2	Dune: Part Three	155
3	KGF Chapter 3	170
4	Salaar 2	165

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-- 7. Language Table

id	name
1	English
2	Hindi
3	Kannada
4	Telugu
5	Tamil

-- 8. Video Formats Table

id	name
1	2D
2	3D
3	IMAX
4	4DX

-- 9. Shows Table

id	screen_id	movie_id	language_id	format_id	audio_format_id	show_time
1	1	1	1	2	1	2026-02-22 10:00:00
2	1	2	1	3	3	2026-02-22 14:00:00
3	1	3	3	1	2	2026-02-22 18:00:00
4	2	1	2	1	1	2026-02-22 11:00:00
5	2	4	4	1	4	2026-02-22 15:00:00
6	3	3	4	1	5	2026-02-22 12:00:00
7	4	1	1	2	2	2026-02-22 10:30:00
8	4	2	2	2	5	2026-02-22 17:00:00

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9	5	4	4	1	3	2026-02-22 13:00:00
10	5	3	3	1	4	2026-02-22 19:00:00
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+						

-- List of all the shows on a given date at a given theatre along with their respective show timings.

brand	branch_name	title	show_time	language	videoFormat	audioFormat	city
PVR Cinemas	Nexus Mall	Avengers: Final War	2026-02-22 10:00:00	English	3D	Dolby Atmos	Bangalore
PVR Cinemas	Nexus Mall	Avengers: Final War	2026-02-22 11:00:00	Hindi	2D	Dolby Atmos	Bangalore
PVR Cinemas	Nexus Mall	Dune: Part Three	2026-02-22 14:00:00	English	IMAX	DTS	Bangalore
PVR Cinemas	Nexus Mall	Salaar 2	2026-02-22 15:00:00	Telugu	2D	Auro 3D	Bangalore
PVR Cinemas	Nexus Mall	KGF Chapter 3	2026-02-22 18:00:00	Kannada	2D	Dolby 7.1	Bangalore